Research Paper 1: HEMS-IoT for Smart-Home Energy Saving

Isaac Machorro-Cano et al. · Energies, 2020

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Overview and Research Problem

The paper *"HEMS-IoT: A Big Data and Machine Learning-Based Smart Home System for Energy Saving"* by Machorro-Cano and colleagues was published in *Energies* in 2020. It addresses the growing electricity demand in residential buildings and the difficulty of reducing energy waste without sacrificing comfort or safety. Smart-home devices can collect and exchange information, but their limited computing and storage resources make it challenging to process large volumes of sensor data directly. The authors therefore combine the Internet of Things (IoT), big-data technologies, machine learning, and home automation in a Home Energy Management System called HEMS-IoT.

The central goal is to monitor household conditions and appliance use, recognize patterns in residents' energy behaviour, classify household energy efficiency, and recommend practical ways to save electricity. The paper emphasizes that effective energy saving is not only a technical problem: residents' habits, comfort preferences, house characteristics, and willingness to act on recommendations all affect the outcome.

System Architecture and Methods

HEMS-IoT is organized as a set of connected layers. The device layer contains household sensors and domotic devices that gather information about the home and its equipment. The communication layer carries data between devices and the system. The management layer processes the information and provides functions such as user management, home management, energy-saving recommendations, and a dashboard displaying indicators and charts. IoT services provide a way to connect with external services, while a security layer and presentation layer support safer operation and interaction with users.

Machine learning is used to identify patterns in energy consumption and user behaviour. The authors use the J48 decision-tree algorithm with the Weka API to learn from household data and classify energy-consumption efficiency. RuleML and Apache Mahout support the generation of energy-saving recommendations that are intended to respect users' preferences, comfort, and safety. Big-data processing helps aggregate and analyze information over different time periods. The design therefore links observation, pattern recognition, classification, and personalized recommendations rather than relying only on fixed rules.

The study includes monitoring data from smart homes and examines consumption at daily, weekly, and monthly scales. Where data were missing, average aggregation was used to produce values at a lower frequency. The analysis considers differences between seasons, time of day, appliance use, and residents' presence or absence. The authors also conducted a user-centred evaluation to assess perceived recommendation quality, satisfaction, usefulness, and intention to use the system.

Why the Approach Matters

A key contribution is the integration of energy monitoring with behaviour-aware recommendations. A dashboard can make consumption visible, while learned patterns can reveal avoidable use—for example, lights or air conditioners operating in empty rooms. Recommendations are meant to help people make better decisions while retaining an acceptable indoor environment. The architecture also anticipates invoking IoT services as connected service providers become available.

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Findings and Evidence

The case study found that air-conditioning systems accounted for approximately 12.5%–22.5% of total energy consumption in the observed context. Consumption was generally higher in the afternoon and at night than in the morning, and it varied by season. The paper reports increased demand during winter associated with heating, and a rise during the July–August school break in some homes that was linked to video-game consoles. In some households, electricity use remained high when residents were absent; excessive lighting and air-conditioning were identified as opportunities for improvement.

The authors report that HEMS-IoT reduced energy consumption by about 42 kWh to 90 kWh every two months in the comparison described in the paper. They attribute the observed savings partly to residents accepting recommendations and changing their home-automation habits. This is important: the system's impact depends not only on its algorithms but also on people following the advice. The paper also reports satisfactory results from a simulation of IoT service requests, although those services were not fully exercised during the household case study.

Conclusions and Limitations

The authors conclude that combining IoT, big-data analytics, and machine learning can support household energy efficiency while considering comfort and safety. J48/Weka is used to learn and classify energy patterns, and RuleML/Mahout supports recommendation generation. The work's main value is its end-to-end design: it collects household information, analyzes usage, identifies inefficient patterns, and turns the analysis into advice for residents.

The evidence should be interpreted in context. The case study and monitoring periods provide practical insight, but they do not establish that the same savings will occur in every home. Household size, climate, appliance mix, building design, and residents' willingness to follow recommendations can change the outcome. The system also depends on reliable sensor data; missing values required aggregation. IoT-service behaviour was evaluated through simulation rather than fully demonstrated in the monitored setting. The paper does not present the work as a universal guarantee of a fixed percentage of savings.

Future Work and Overall Assessment

The authors propose deploying HEMS-IoT at a larger scale and across homes with more domotic devices. They also suggest location-based functions using mobile-device GPS, so the system could estimate when residents will return and prepare the home for comfort. Adding solar panels is another planned direction for reducing grid electricity use. The authors further identify stronger security measures, including blockchain and cybersecurity techniques, as future work.

Overall, this paper presents a user-oriented smart-home energy management architecture. Its strength is connecting data analytics with personalized recommendations and evaluating both consumption patterns and user perceptions. Its central lesson is that energy-saving technology works best when accurate monitoring, useful advice, resident behaviour, comfort, and security are considered together.

Research Paper 2: Reinforcement Learning for Home Energy Management

Sangyoon Lee and Dae-Hyun Choi · Sensors, 2019

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Overview and Objective

The paper *"Reinforcement Learning-Based Energy Management of Smart Home with Rooftop Solar Photovoltaic System, Energy Storage System, and Home Appliances"* by Sangyoon Lee and Dae-Hyun Choi was published in *Sensors* in 2019. It studies how a home energy management system (HEMS) can schedule electricity use to lower a household's bill while maintaining indoor comfort and respecting appliance operating requirements. The system includes rooftop solar photovoltaic (PV) generation, an energy storage system (ESS), an air conditioner, a washing machine, and other household loads.

Traditional optimization approaches, such as mixed-integer linear programming (MILP), can produce schedules but depend on a mathematical model of the system. The authors propose a data-driven, model-free reinforcement-learning approach. Their main contribution is to apply Q-learning to appliance scheduling and ESS charging/discharging, and to combine it with an artificial neural network (ANN) that predicts indoor temperature for improved air-conditioner control.

System Model and Learning Method

The HEMS operates under time-of-use (TOU) electricity prices, where electricity costs vary by time. It considers different load types. A controllable or reducible load, such as an air conditioner, can adjust its energy use. A shiftable load, such as a washing machine, can be moved to a suitable time window but may need to run for a continuous cycle. Uncontrollable appliances are treated as fixed loads. The ESS can charge or discharge within its operating limits, while PV generation offsets some household demand.

The objective balances two goals: reducing electricity cost and limiting consumer discomfort. For the air conditioner, comfort is represented through the preferred indoor-temperature range. For the washing machine, the schedule must respect the user's preferred operating period and its consecutive operating duration. For the ESS, charging and discharging decisions must avoid overcharging or undercharging.

The proposed method uses separate Q-learning agents for the air conditioner, washing machine, and ESS. Each agent observes its state, selects an action, and receives a reward that reflects electricity cost and relevant penalties. Through repeated interaction with the environment, the agents learn policies intended to maximize cumulative reward. The washing-machine agent chooses when the cycle should run; the ESS agent determines charging and discharging; and the air-conditioner agent selects energy use while considering temperature comfort.

An ANN is trained to approximate the relationship between indoor temperature, outdoor conditions, and air-conditioner energy consumption. This prediction is incorporated into the air-conditioner agent's learning process. The study evaluates the approach through computer simulation and compares it with a conventional MILP-based HEMS.

Simulation Setup

The simulated household is scheduled over 24 hours with one-hour time slots. The model includes an air conditioner, washing machine, ESS, PV generation, and time-varying electricity prices. The assumed preferred indoor temperature range is 23–25 °C, with a preferred set point of 24 °C. The washing machine must operate for two consecutive hours within a permitted daytime/evening window. The ESS has specified minimum and maximum energy levels and charging/discharging limits. The simulations were implemented using MATLAB for MILP and Python for Q-learning and the ANN.

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Results

The headline result is that the proposed Q-learning and ANN-based HEMS reduces the electricity bill by 14% relative to the existing optimization approach used for comparison. The paper reports that the algorithm can schedule the air conditioner and washing machine and manage ESS charging/discharging while maintaining the preferred comfort level and appliance operating characteristics. The result is obtained in simulation for a single household; it is not a field trial across many homes.

The individual agents learn different but coordinated practical behaviours. The washing-machine agent schedules its cycle within the user's preferred period. The ESS agent decides when to store energy and when to discharge it, while respecting its operating boundaries. The air-conditioner agent uses the ANN-based indoor-temperature prediction to make decisions that balance energy cost against thermal comfort. Together, these decisions can use solar generation and storage more effectively and shift demand away from expensive time slots where feasible.

The study also examines the effect of reward-function penalty parameters. These parameters express the relative importance of discomfort, appliance timing, and ESS constraints. The paper's analysis indicates that the trade-off matters: prioritizing comfort more strongly can reduce discomfort but may limit cost savings. The ANN is evaluated for its contribution to the air-conditioner agent by improving its representation of the relationship between temperature and energy use.

Conclusions and Significance

The authors conclude that reinforcement learning combined with an ANN is a promising alternative to model-based scheduling for smart-home energy management. Its advantage is that agents can learn policies through interaction without requiring a fully specified system model for every decision. The study's 14% relative bill reduction demonstrates potential under the simulation assumptions, while the explicit comfort and appliance constraints make the approach more realistic than cost-only scheduling.

However, the findings have limitations. The experiment represents one simulated home, a fixed 24-hour horizon, and assumed PV generation and outdoor-temperature forecasts. Performance in real households could be affected by forecast errors, changing occupancy, different tariffs, battery degradation, device constraints, and communication reliability. A simulated cost reduction should therefore not be interpreted as a guaranteed saving for every user. Practical testing is needed to determine how the learned policy behaves under real operating uncertainty.

Future Work and Overall Assessment

The authors propose extending the approach to multiple homes with distributed energy resources and smart appliances. This would require an efficient communication strategy between homes while preserving energy savings and user comfort. They also recommend testing the method in larger, realistic electric-power networks. A further direction is to use more advanced neural-network models, including recurrent neural networks and long short-term memory (LSTM), to improve indoor-temperature prediction.

In summary, this paper focuses on control and scheduling rather than energy prediction alone. Q-learning makes decisions about appliance timing and storage operation, while the ANN supplies a useful temperature prediction for air-conditioning control. Its key contribution is a combined strategy that considers electricity price, renewable generation, storage, appliance constraints, and user comfort within one HEMS framework.

Research Paper 3: LSTM-Based Sustainable Smart-Home Energy Management

Muhammad Adnan Khan et al. · Discover Sustainability, 2024

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Overview and Research Objective

The paper *"Optimizing Smart Home Energy Management for Sustainability Using Machine Learning Techniques"* by Muhammad Adnan Khan and colleagues was published in *Discover Sustainability* in 2024. It investigates machine-learning-based prediction of household energy consumption as a way to support more sustainable smart-home energy management. As smart homes add connected appliances and monitoring systems, the amount and variability of energy data increase. Accurate forecasts can help a home energy management system understand usage patterns and make better-informed energy decisions.

The study proposes a multivariate Long Short-Term Memory (LSTM) model. LSTM is a type of recurrent neural network designed to learn patterns across sequences, making it suitable for data whose current values are related to earlier observations. The authors compare prediction performance using error measures and report that the multivariate LSTM performs better than the previous approaches considered in their evaluation.

Data and Methodology

The proposed Energy Consumption Model is organized into input, preprocessing, training, performance evaluation, and validation stages. Energy sensors gather readings from household devices, and the data are passed through preprocessing before model training. The paper describes one-minute readings over a 350-day period, recording electricity use in kilowatts for different household appliances through smart meters. This provides time-series information from which the model can learn changing usage patterns.

Preprocessing includes normalization and moving averages to handle missing values and bring data into a suitable range. The dataset is split into 70% for training and 30% for validation/testing, as described in the paper. The training stage learns the model from the training data. A performance stage evaluates its errors and accuracy; if the desired criteria are not met, the training process is adjusted. The trained model is then stored and used in the validation stage to generate predictions on held-out data.

The research evaluates the LSTM multivariate model using several metrics. Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) measure prediction error, with smaller values generally indicating closer predictions. Mean Absolute Error (MAE) measures average absolute error. Mean Absolute Percentage Error (MAPE) expresses error as a percentage, although its interpretation can depend on the scale and values in the data. Mean Absolute Scaled Error (MASE) compares performance with a simple baseline forecast. The coefficient of determination (R^2) indicates how much variation in the observed values is accounted for by the model.

Connection to Sustainability

The proposed approach is primarily a forecasting method. Rather than directly switching appliances or controlling batteries, it aims to estimate consumption accurately so that a HEMS can use those predictions in later planning. Forecasting can support energy-awareness tools, load planning, and more informed decisions about demand. The paper situates this work within a broader smart-home context involving smart meters, IoT communication, cloud computing, connected appliances, and the integration of renewable generation and storage.

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Results and Model Performance

The authors identify the multivariate LSTM as the strongest model in their evaluation and report the following metrics: MSE = 0.02284, RMSE = 0.15113, MAE = 0.184, MAPE = 0.123, MASE = 0.996, and R^2 = 0.694. These figures are reproduced as reported in the paper. The low MSE and RMSE are presented by the authors as evidence of relatively small forecasting errors, while the R^2 value indicates that the model explains a meaningful portion—but not all—of the variation in energy consumption.

The abstract and conclusion describe the model as reliable and useful for smart, sustainable energy management. At the same time, metrics should be interpreted together rather than treating any single value as proof of perfect prediction. The reported R^2 of 0.694 means that substantial variation remains unexplained. MASE is close to 1, so its value should be read in relation to the baseline and the precise evaluation setup. The paper reports the metrics but the practical benefit of the forecasts depends on how they are integrated into real energy-management decisions.

Conclusions and Implications

The paper concludes that machine learning, especially multivariate LSTM, can be used to predict smart-home energy consumption and support better energy management. Its workflow—sensor data collection, preprocessing, model training, error evaluation, and validation—provides a structured approach for developing a forecasting component for HEMS. Better forecasts could help identify expected demand and support decisions intended to improve energy efficiency and sustainability.

A useful distinction is that this paper predicts energy consumption, whereas a control-oriented system such as reinforcement-learning HEMS uses decisions to schedule appliances and storage. Prediction and control can complement each other: forecasts can inform a controller, and a controller can act on predicted demand, renewable generation, and electricity prices. The 2024 study contributes to the prediction side of that larger system.

Limitations and Overall Assessment

The reported results are based on the dataset and evaluation design described by the authors. They do not, by themselves, prove that the model will achieve the same accuracy in different homes, climates, appliance mixes, or future periods. The reported R^2 also shows that some consumption variation is not captured. Real-world deployment would need to consider sensor quality, missing or noisy readings, changes in residents' behaviour, and whether the training data remain representative over time. The article's reported metrics should not be converted directly into a percentage of electricity-bill savings because forecasting accuracy and actual cost reduction are different outcomes.

Overall, this research supports the use of time-series machine learning as one component of sustainable smart-home management. Its main contribution is the application and evaluation of a multivariate LSTM forecasting model, supported by a staged data-processing and validation workflow. Future practical value will depend on robust testing and on connecting predictions to user-aware energy-management strategies.